PROJECT MILESTONE 2
CMSC 170

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1. Data Preprocessing Plan

The dataset used for this project is the Conflict Data for Philippines dataset from the Uppsala Conflict Data Program (UCDP), hosted on the Humanitarian Data Exchange (HDX). The main goal is to predict whether a recorded conflict involved the CPP/NPA or a Mindanao regional armed group using information available in the dataset.

The target variable is side_b, which identifies the rebel or insurgent group involved in a conflict. The input features are longitude, latitude, deaths_a, deaths_b, deaths_civilians, best, type_of_violence, and duration_days.

The data loading and preprocessing code is separated from the implementations of the machine learning algorithms. In the current program, main.py calls the preprocess() function from preprocess.py to load and preprocess the CSV file named conflict_data_phl.csv. The processed data is then passed to the dataset splitting function in split.py.

The preprocessing function removes records belonging to three excluded categories: Civilians, Government of Kuwait, and Military faction (forces of Honasan, Abenina & Zumel). It also calculates the duration of each conflict in days using the start and end dates. If either date is missing, the duration is set to 0.

The processed records follow this order: longitude, latitude, deaths_a, deaths_b, deaths_civilians, best, type_of_violence, duration_days, and side_b. The numerical features are converted into their appropriate numerical types, while side_b is kept as the original armed-group label. The target variable is placed at the last index of each record, allowing the splitting code to group records according to their side_b values.

After the dataset is divided into training and testing sets, the classify_groups() function converts the original armed-group labels into two classes. CPP is assigned the value 0, while all other included groups are assigned the value 1. This means the project performs binary classification between CPP/NPA and Mindanao regional groups rather than predicting the individual armed group responsible for each conflict.

2. Train/Test Split Plan

The project uses an 80/20 train/test split. Approximately 80% of the records will be used to train the models, while the remaining 20% will be reserved for testing their predictions. This allows

the models to learn from one portion of the dataset and be evaluated using records that were not included in their training data.

The dataset is initially divided according to the original target variable side_b. In split.py, the program first creates a dictionary in which each key represents an original armed group and its corresponding value contains all records associated with that group. This separates the dataset into groups before the training and testing sets are created.

For each armed group, the program calculates the number of records to include in the training set by multiplying the group's total number of records by 0.8 and converting the result to an integer. Before dividing the records, the program randomly shuffles the records within each group. The first 80% of the shuffled records are added to the training set, while the remaining records are added to the testing set. The process is repeated for every group, and the resulting records are combined into the final training and testing sets.

The program uses a fixed random seed of 67. This makes the random splitting process reproducible, meaning the same dataset will produce the same split when the program is run again. After combining the records from all groups, the program also shuffles the final training and testing sets.

This approach aims to maintain a roughly proportional distribution of the original armed groups in the training and testing sets. It is particularly useful for this dataset because some groups have many more recorded conflicts than others. Separating the records by side_b before splitting helps ensure that the division is performed within each original group instead of dividing the entire dataset without considering its target classes.

After the splitting process, the original armed-group labels are converted into two binary classes. CPP is assigned 0, while all other included groups are assigned 1. This means the project ultimately classifies conflicts as either CPP/NPA clashes or Mindanao regional group clashes.

The program uses the same training and testing sets for K-Nearest Neighbor, Gaussian Naive Bayes, and Perceptron so that their performance can be compared under the same conditions. The training set will be used to develop each model, while the testing set will be used to evaluate its predictions.

3. Evaluation Metrics

The project will use Weighted F1-Score, a Binary Confusion Matrix, and Balanced Accuracy (Macro-Averaged Recall) to evaluate the performance of the three machine learning methods. These metrics are appropriate for examining predictions across the two classes, especially because the number of recorded incidents may differ between CPP/NPA and Mindanao regional groups.

3.1 Weighted F1-Score

Weighted F1-Score will serve as the main evaluation metric. It combines precision and recall into one measure. Precision indicates how often a model's predictions for a particular class are correct, while recall indicates how many of that class's actual incidents the model successfully identifies.

The F1-Score is calculated using the following formula:

F1-Score = 2 × (Precision × Recall) / (Precision + Recall)

The weighted version calculates the F1-Score for each class (CPP/NPA vs. Mindanao regional groups) and combines the results according to the number of actual records belonging to each class. This provides an overall measure of performance that accounts for the different numbers of recorded incidents between the two classes.

This metric is useful if the dataset is imbalanced, meaning one class contains significantly more recorded incidents than the other. It allows the evaluation to consider the performance of both classes while accounting for their different numbers of samples.

3.2 Binary Confusion Matrix

A Binary Confusion Matrix will be used to examine the specific prediction errors made by each model. It compares the actual classes with the classes predicted by the model. In this project, the two classes are CPP/NPA clashes and Mindanao regional group clashes.

The matrix contains four possible results: True Positives, False Positives, True Negatives, and False Negatives. These values show how many predictions were correct and how many were incorrect for each class.

The confusion matrix will help identify whether the model incorrectly classifies a CPP/NPA clash as a Mindanao regional group clash, or incorrectly classifies a Mindanao regional group clash as a CPP/NPA clash. These errors are important because they show which type of conflict the model has difficulty identifying.

Correct predictions appear along the main diagonal of the matrix, while incorrect predictions appear outside the diagonal. Examining these values will provide a clearer understanding of the models' strengths and weaknesses when distinguishing between the two classes.

3.3 Balanced Accuracy (Macro-Averaged Recall)

Balanced Accuracy will be used to evaluate how well the models identify both classes, regardless of how many records each class contains.

In binary classification, Balanced Accuracy is calculated by taking the average of the recall for CPP/NPA clashes and the recall for Mindanao regional group clashes. It can also be understood as the average of sensitivity and specificity.

Balanced Accuracy = (Recall_CPP/NPA + Recall_Mindanao) / 2

This metric is important because a model might perform well on the class with more recorded incidents while failing to correctly identify the class with fewer records. Balanced Accuracy gives both classes equal importance, helping determine whether the models can recognize both types of clashes instead of mainly predicting the majority class.

The three metrics will be applied to predictions on the same testing set for all three methods. Their results will then be compared to determine their overall performance and ability to distinguish between CPP/NPA clashes and Mindanao regional group clashes.

4. Chosen Methods

4.1 K-Nearest Neighbor (KNN)

K-Nearest Neighbor will predict whether a conflict belongs to the CPP/NPA class or the Mindanao regional groups class by comparing an incident with similar incidents in the training dataset. It identifies the nearest records based on the selected input features and predicts the class that appears most frequently among those neighbors.

The current implementation uses 5 neighbors (k = 5) and a Minkowski distance with p = 2, which is equivalent to Euclidean distance. It calculates the distance using all eight numerical input features and uses majority voting to determine the predicted class.

This method was selected because geographical information, casualty counts, and other conflict details may help distinguish between the two classes. Longitude and latitude may help the model identify patterns based on where previous conflicts occurred.

4.2 Gaussian Naive Bayes

Gaussian Naive Bayes will predict whether a conflict belongs to the CPP/NPA class or the Mindanao regional groups class by estimating the probability that an incident belongs to each class based on its input features. It assumes that numerical features follow a Gaussian distribution within each class and that the features are conditionally independent given the class.

The current implementation calculates the prior probability, mean, and variance of each feature for both classes. It then uses Gaussian log likelihoods to estimate which class is more likely to contain the given incident. A small smoothing value of 1e-9 is added to the calculated variances to help avoid problems when a feature has very little or no variation.

This method was selected because the dataset contains numerical information, including geographical coordinates, casualty counts, and conflict duration. Differences in these values may help distinguish between CPP/NPA clashes and Mindanao regional group clashes.

4.3 Perceptron

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The Perceptron will be used as a linear classification method that learns weights for the input features and uses them to predict whether a conflict belongs to the CPP/NPA class or the Mindanao regional groups class. During training, the model adjusts its weights and bias when its predictions are incorrect.

The current implementation uses a learning rate of 0.01 and a maximum of 100 training epochs. It initializes the weights and bias to zero and uses a sign function to determine the predicted class. The model uses the two binary classes by assigning one class a target value of 1 and the other a target value of -1 during training.

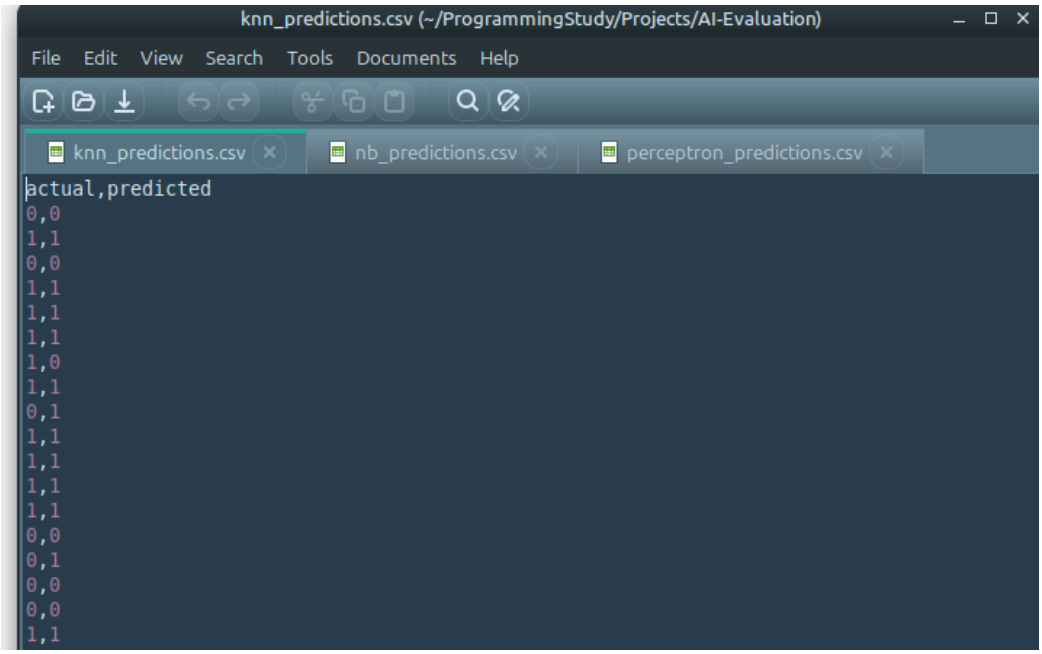
This method was selected to determine whether a relatively simple linear model can distinguish between the two classes using geographical information, casualty counts, and other conflict details.

All three methods are implemented separately and use the same preprocessed dataset and training and testing sets. Their predictions are saved into separate CSV files: nb_predictions.csv, perceptron_predictions.csv, and knn_predictions.csv. These files contain the actual and predicted labels for each testing record, allowing the results of the three methods to be compared.

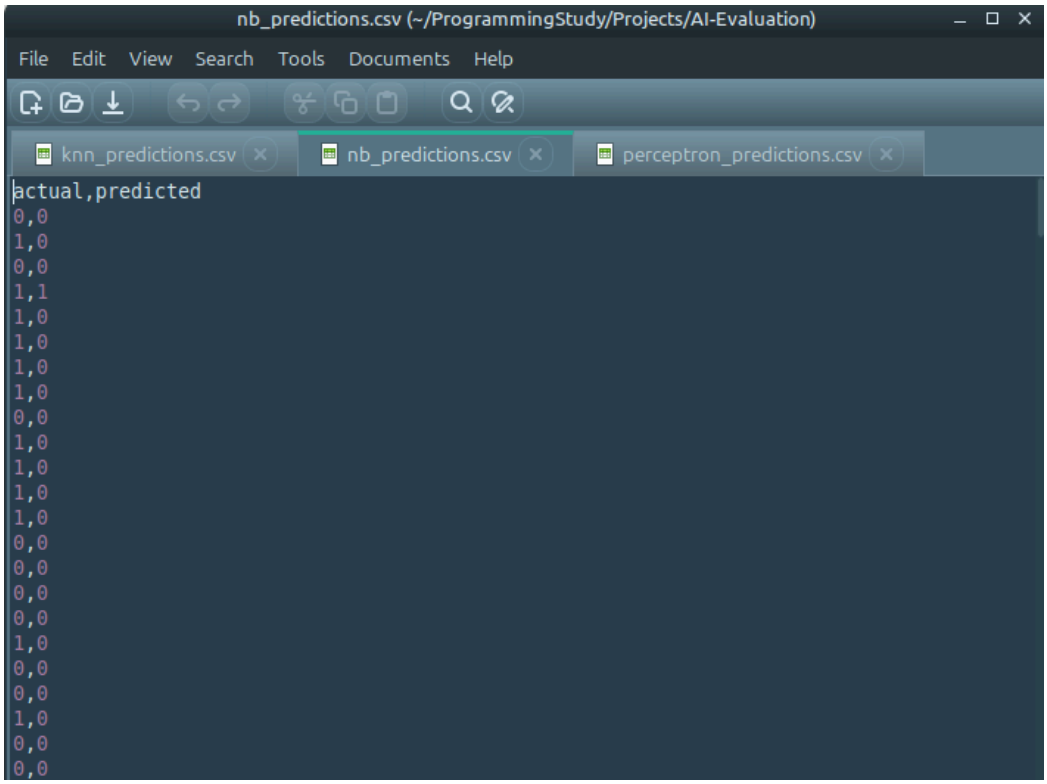
The current program saves the predictions but does not yet calculate the Weighted F1-Score, Binary Confusion Matrix, or Balanced Accuracy automatically. These metrics will need to be calculated separately using the saved predictions.

Initial Results:

1. **KNN**



2. Gaussian Naive Bayes



3. Perceptron

